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SIMATS ENGINEERING

ASSESSMENT TOOL 2

Case study

COURSE NAME: Computer Networks **COURSE CODE:** CSA0720

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Case study

83%

QUESTIONS:4 **Total Marks:40**

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss, 180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

Questions

1. Analyze whether the problem is primarily caused by the **transport protocol (TCP/UDP)** or the **application layer**.
2. Justify your answer based on the communication requirements of real-time multimedia applications.
3. Recommend suitable protocol-level or application-level improvements to enhance conferencing quality.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization

1.A multinational IT company uses a cloud-based video conferencing platform for daily client meetings. During peak business hours, employees experience frequent voice distortion, frozen video frames, and delayed communication. Network monitoring reports indicate an **8% packet loss**, **180 ms jitter**, and fluctuating bandwidth utilization. The IT administrator suspects that the issue is related to the transport protocol or application-level buffering strategy.

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ANSWER:

1. Cause

The problem is mainly related to **UDP and application-level buffering**. Real-time video conferencing commonly uses UDP because low delay is more important than retransmitting every lost packet.

The observed **8% packet loss and 180 ms jitter** are significant for real-time communication. UDP does not retransmit lost packets, so missing packets may directly affect audio and video quality. The application must therefore handle packet loss and timing variations efficiently.

2. Justification

- **8% packet loss** → causes missing audio/video data, resulting in broken voice and incomplete video frames.
- **180 ms jitter** → causes irregular packet arrival, leading to voice distortion and video instability.
- **Fluctuating bandwidth** → causes frozen video and sudden reductions in video quality.
- **TCP retransmission** can increase delay, making real-time communication worse.
- Real-time applications require **low latency and continuous data delivery** rather than perfect retransmission of every packet.
- Excessive buffering can also increase delay between the speakers, making conversations feel unnatural.
- Therefore, UDP combined with an intelligent application-level buffering and congestion-control strategy is more suitable.

3. Improvements

- Use **adaptive bitrate** to automatically reduce video quality when bandwidth decreases.
- Use **jitter buffers** to handle variations in packet arrival time.
- Apply **QoS** to prioritize voice and video traffic over less time-sensitive applications.
- Use efficient video/audio **codecs** to reduce bandwidth requirements.
- Use congestion-control mechanisms suitable for real-time traffic.
- Increase available network bandwidth where possible.
- Monitor packet loss and jitter continuously.
- Reduce unnecessary network traffic during peak business hours.
- Use geographically closer conferencing servers to reduce network delay.

2.A multinational software company hosts its web services across multiple continents. Employees frequently report long delays before the application homepage loads. Network analysis shows that the **average DNS lookup time is approximately 800 ms**, significantly affecting user experience. The organization plans to redesign its DNS infrastructure while ensuring continuous service availability during server failures.

Questions

1. Analyze the reasons behind the high DNS resolution time.
2. Propose a suitable DNS architecture using techniques such as **Anycast DNS**, **DNS pre-fetching**, and **TTL optimization** to reduce the lookup time below **50 ms**.
3. Evaluate the advantages and limitations of your proposed architecture in terms of performance, scalability, and availability.

ANSWER:

1. Reasons for 800 ms DNS Resolution

The DNS resolution time is approximately **800 ms**, which is very high for a frequently accessed web application.

Possible reasons include:

- Requests are directed to **distant DNS servers**.
- A single-server dependency creates congestion.
- Poor DNS caching increases repeated lookups.
- High network latency increases response time.
- DNS server overload can delay query processing.
- Network congestion between users and the DNS server can increase lookup time.
- Failure or slow response from the primary DNS server can further increase delays.
- Repeated queries for frequently accessed domains unnecessarily consume DNS resources.

2. Proposed Architecture

Use:

Anycast DNS + DNS Caching + Prefetching + Optimized TTL

Architecture:

Users
↓
Nearest Anycast DNS Server
↓
DNS Cache
↓
Authoritative DNS
↓
Application Server

Implementation:

- Deploy DNS servers in **multiple geographic locations**.
- Use **Anycast** to direct users to the nearest available DNS server.
- Use DNS caching to avoid repeated lookups.
- Use prefetching for frequently accessed domains.
- Optimize TTL values to balance freshness and caching efficiency.
- Use redundant DNS servers so that failure of one server does not interrupt resolution.
- Monitor DNS response time continuously.

Target resolution time: Below **50 ms** for users served by nearby DNS infrastructure.

3. Advantages and Limitations

Advantages

- Lower DNS latency.
- High availability.
- Better scalability.
- Automatic traffic distribution.
- Reduced load on individual DNS servers.
- Faster response through caching.
- Failure of one server does not stop DNS service.
- Better performance for users in different geographic regions.

Limitations

- Higher infrastructure cost.
- More complex management.
- Requires multiple DNS locations.
- Incorrect TTL settings may cause stale records.
- DNS configuration and monitoring become more complicated.
- Anycast routing requires proper network configuration.

3.A cloud service provider delivers applications to millions of users worldwide. During promotional events, users experience intermittent delays in accessing cloud-hosted services. Investigation reveals that DNS queries are taking nearly **800 ms** because all requests are directed to a single primary DNS server. The company wants a highly available DNS solution capable of handling global traffic efficiently.

Questions

4. Examine the impact of centralized DNS architecture on application performance.
5. Design a distributed DNS solution that minimizes lookup latency while maintaining high availability.
6. Analyze how **TTL values**, **Anycast routing**, and **DNS caching** influence system performance and fault tolerance.

ANSWER:

4. Impact of Centralized DNS

A single DNS server becomes both a **performance bottleneck and a single point of failure**.

The 800 ms lookup delay directly increases the time required to access cloud applications. Even if the cloud application itself responds quickly, users experience a delay before the connection begins.

Other effects include:

- Increased DNS server workload.
- Poor performance during promotional events.
- Higher probability of service interruption.
- Increased latency for geographically distant users.
- Limited scalability when the number of users increases.

- Failure of the primary DNS server can affect all users.

5. Proposed Solution

Use a distributed DNS architecture:

```
           Users
             ↓
         Anycast DNS
             ↓
   ┌─────────────────┼───────────────┐
   ↓          ↓        ↓
 DNS-1            DNS-2      DNS-3
 Region A         Region B   Region C
            ↓
         DNS Cache
            ↓
       Cloud Application
```

Deploy multiple DNS servers across different geographic regions and use **Anycast routing**.

The nearest available DNS server responds to the user's request. Cached responses can be returned immediately without contacting the authoritative server for every request.

6. Role of TTL, Anycast and Caching

TTL

TTL controls how long DNS records remain cached.

- Appropriate TTL reduces repeated DNS queries.
- Longer TTL can improve performance for stable records.
- Very long TTL values can delay updates.

Anycast

Anycast allows the same DNS service address to be announced from multiple locations.

- Users are directed toward a nearby DNS server.
- Reduces network latency.
- Provides redundancy if one location becomes unavailable.

Caching

Caching stores previously resolved DNS records.

- Reduces repeated queries.
- Reduces DNS server load.
- Improves response time.
- Helps handle sudden traffic increases.

Together, TTL optimization, Anycast routing, and DNS caching improve performance, scalability, availability, and fault tolerance.

4.A financial institution operates an electronic stock trading platform where thousands of buy and sell orders are submitted every second. During market opening hours, the **average order acknowledgment latency increases from 1 ms**

to 40 ms, resulting in delayed trade confirmations and customer dissatisfaction. Network engineers observe increased TCP retransmissions, longer connection establishment times, and excessive buffering.

Questions

1. Analyze three possible TCP-related factors responsible for the latency increase, considering **flow control**, **Nagle's algorithm**, and **SYN queue limitations**.
2. Explain how each factor affects transaction processing in high-frequency systems
3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

ANSWER:

1. Three TCP Factors

a) Flow Control

TCP receive windows control how much data can be sent before acknowledgments are received.

- Small receive windows can limit the amount of outstanding data.
- Frequent pauses can reduce throughput.
- Buffer limitations can become more visible during high transaction rates.
- Properly sized TCP buffers can help maintain continuous data transmission.

b) Nagle's Algorithm

Nagle's algorithm combines small TCP packets before transmission.

- It reduces the number of small packets on the network.
- However, it can introduce additional delay.
- In financial trading, messages are often small and latency-sensitive.
- Therefore, waiting to combine packets can negatively affect order-response time.
- **TCP_NODELAY** can be considered for latency-sensitive traffic.

c) SYN Queue Limitations

The SYN queue stores connection requests while TCP connections are being established.

- A limited SYN queue can delay new connections.
- During market opening, many simultaneous connections may arrive.
- Queue exhaustion can result in delayed or failed connection establishment.
- Increasing and properly tuning the SYN backlog can help handle connection bursts.

2. Effect on Transactions

These factors directly affect transaction processing:

- **Flow control** → delays data transmission and may reduce throughput.
- **Nagle's algorithm** → increases latency for small trading messages.
- **SYN queue limitations** → increase connection establishment time.
- **TCP retransmissions** → cause additional delays when packets are lost.
- **Excessive buffering** → can increase queueing delay.
- Longer connection establishment can delay the submission and acknowledgment of orders.

These factors can contribute to order acknowledgment latency increasing from **1 ms to 40 ms** during peak trading periods.

3. Recommended Changes

- Increase TCP receive/send buffer sizes where appropriate.
- Disable Nagle's algorithm using **TCP_NODELAY** for latency-sensitive trading traffic.
- Increase and properly tune the **SYN backlog/queue**.
- Use persistent TCP connections where possible to avoid repeated handshakes.
- Monitor TCP retransmissions continuously.
- Monitor network congestion and packet loss.
- Ensure sufficient network bandwidth during market-opening periods.
- Reduce unnecessary buffering where low latency is more important than throughput.
- Use dedicated or optimized network paths for critical trading traffic.
- Continuously monitor order acknowledgment latency.

3. Recommend appropriate TCP configuration changes to reduce latency and improve transaction reliability.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand

MARK ALLOCATION

	Understanding of Case (25%)	Application of Concepts (25%)	Analysis (20%)	Solution Design (20%)	Clarity (10%)	
Q1	2	2	2	2	1	9
Q2	2	2	2	2	0.5	8.5
Q3	2.5	1.5	1.5	1	0.5	7
Q4	2.5	2	1.5	1.5	1	8.5
~~Q5~~	~~2~~	~~2.5~~	~~2~~	~~2~~	~~1~~	~~9~~

~~42~~

33/40